

Example 5.20. Determine the peak output voltage for a positive series clipper circuit shown in Fig. 5.30. The input signal is sinusoidal of peak value 15 V and the barrier voltage for silicon diode is of 0.7 V.

Solution: During the positive half cycle of the input signal, the diode D remains reverse biased and so does not conduct. Thus output voltage remains zero throughout the positive half cycle.

During the negative half cycle of the input signal, the diode remains forward biased and so conducts heavily. As a result the output voltage (peak value) will be input voltage (peak value) less diode barrier voltage. i.e.,

$$V_{out(peak)} = V_{in(peak)} - V_0 = 15 - 0.7 = 14.3 \text{ V Ans.}$$

Example 5.21. What is clipper circuit? Sketch the output voltage waveform for the circuit shown in Fig. 5.31 (a).

[G.B. Technical Univ. Electronics Engineering First Semester 2010-11]

Solution: During positive half cycle of the input signal, the diode remains forward biased and the current flowing through the circuit is

$$\frac{V_{in} - 0.7}{(2.2 + 3.3) \text{ k}\Omega} = \frac{10 - 0.7}{5.5 \text{ k}\Omega} = 1.691 \text{ mA}$$

$$\text{and Output voltage, } V_{out} = 1.691 \times 10^{-3} \times 3.3 \times 10^3 = 5.58 \text{ V}$$

During negative half cycle of the input signal, the diode remains reverse biased and so does not conduct. The output voltage remains zero.

Output voltage waveform is shown in Fig. 5.31 (b).

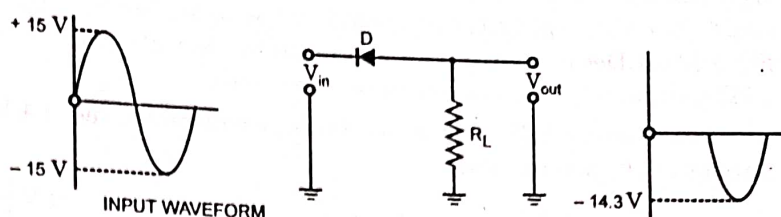


Fig. 5.30

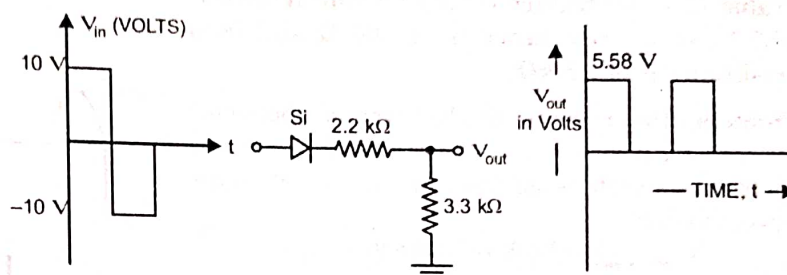


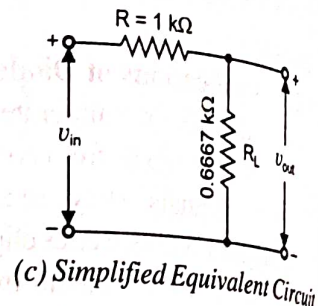
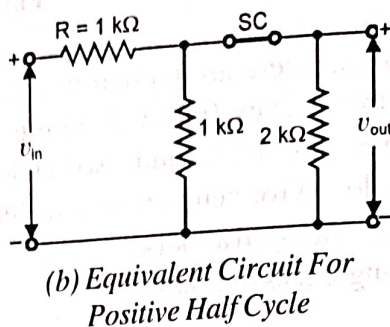
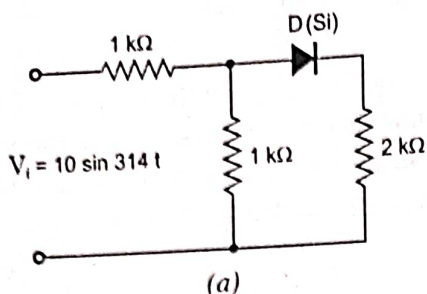
Fig. 5.31 (a)

Fig. 5.31 (b)

Example 5.22. Find the voltage drop across $2\text{ k}\Omega$ resistance in the circuit shown in Fig. 5.32 (a) (Assume diode to be ideal).

Solution: Given that the diode is ideal. During the positive half cycle of the input sinusoidal signal, the diode is forward biased and is replaced by a short circuit, as shown in Fig. 5.32 (b), because the diode is ideal. This equivalent circuit is further simplified as shown in Fig. 5.32 (c) because load resistance

$$R_L = 1\text{ k}\Omega \parallel 2\text{ k}\Omega = \frac{1 \times 2}{1 + 2} = 0.6667\text{ k}\Omega$$



Under this condition, $V_{\text{out (peak)}}$

= Peak voltage across $0.6667\text{ k}\Omega$ resistor

$$= V_{\text{in (peak)}} \times \frac{R_L}{R + R_L} = 10 \times \frac{0.6667}{1 + 0.6667} = 4\text{ V}$$

Output voltage waveform with peak value of 4 V is shown in Fig. 5.32 (d).

During the negative half cycle of the input sinusoidal signal, the diode is reverse biased and so it does not conduct and, therefore, is replaced by an open circuit as shown in Fig. 5.32 (e). Due to open circuit, no current flows through $2\text{ k}\Omega$ resistor and, therefore, output voltage is zero.

Thus output is half-wave rectified voltage with peak value of 4 V . So output voltage (rms value)

$$= \frac{V_{\text{out (peak)}}}{2} = \frac{4}{2} = 2\text{ V Ans.}$$

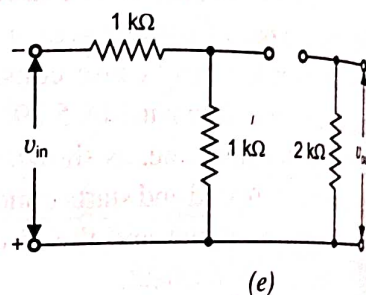
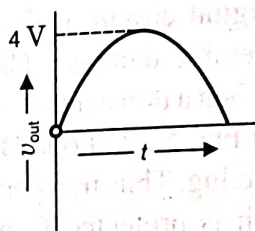


Fig. 5.32

Example 5.23. Determine the peak output voltage for a negative shunt clipper circuit shown in Fig. 5.33 (a). The input signal is sinusoidal of peak value 12 V , the barrier voltage for silicon diode is 0.7 V , series resistance is of $500\ \Omega$ and load resistance is of $2.5\text{ k}\Omega$.

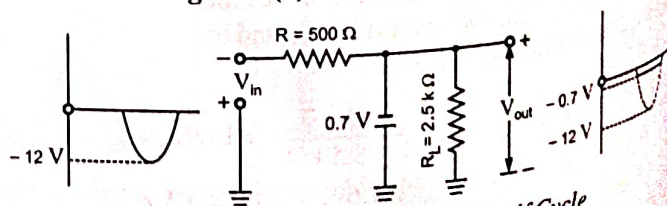
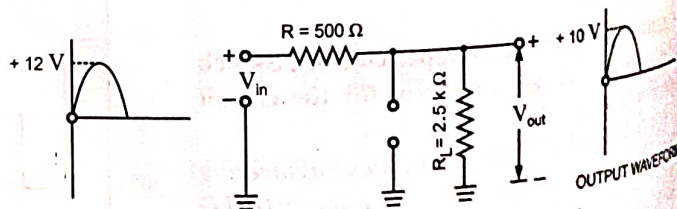
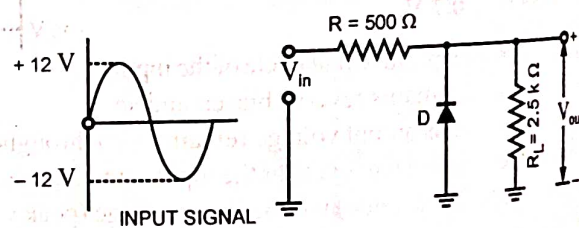
Solution: During the positive half cycle of input signal voltage, the diode remains reverse biased and behaves as an open switch, as illustrated in Fig. 5.33 (b). Under this condition

$V_{\text{out (peak)}}$ = Peak voltage across R_L =

$$V_{\text{in (peak)}} \times \frac{R_L}{R + R_L} = 12 \times \frac{2,500}{500 + 2,500} = 10\text{ V}$$

During the negative half cycle of the input sinusoidal signal, the diode remains forward biased. The diode starts conducting when applied voltage exceeds -0.7 V . Since diode D and load resistor R_L are in parallel, voltage across them cannot exceed -0.7 V . Thus negative half cycle beyond -0.7 V is clipped.

Thus output voltage during negative half cycle is given as $V_{\text{out}} = -0.7\text{ V}$.



Example 5.24. An input of $E = \pm 50$ V is applied to the clipping circuit. The output current from the circuit is to be $I_L = 20$ mA and negative output voltage is not to exceed -0.5 V.

Calculate the value of R_1 . Specify the diode in terms of forward current, power dissipation and peak reverse voltage. Assume the saturation current $I_S = 5 \mu\text{A}$ and forward voltage of diode, $V_F = 0.7$ V.

[U.P.S.C. I.E.S. Electrical Engineering-II, 2011]

Solution:

Reverse saturation current, $I_S = 5 \mu\text{A}$

Negative output voltage, $V_{\text{out}} \leq -0.5$ V

Output will be negative when input voltage, $E = -50$ V

and the diode D will be reverse biased then current through resistor R_1 ,
 $I = I_S = 5 \mu\text{A} = 5 \times 10^{-6}$ A

Output voltage, $V_{\text{out}} \leq -0.5$ V taking magnitude only

$$\text{So } I_S R_1 = V_{\text{out}} \leq 0.5 \text{ V}$$

$$\text{or } R_1 \leq \frac{0.5}{I_S} \leq \frac{0.5}{5 \times 10^{-6}} \leq 100 \text{ k}\Omega$$

$$\text{Let } R_1 = 100 \text{ k}\Omega \text{ Ans.}$$

$$\text{Forward current through diode, } I_F = \frac{E - V_F}{R_1} + I_L = \frac{50 - 0.7}{100 \times 10^3} + 20$$

$$= 20.493 \text{ mA Ans.}$$

$$\text{Power dissipation} = V_F I_F$$

$$= 0.7 \times 20.493 \times 10^{-3} = 14.345 \text{ mW Ans.}$$

$$\text{Peak inverse voltage, } V_P = E - 0.5 = 50 - 0.5 = 49.5 \text{ V Ans.}$$

$\therefore 50$ V is maximum -ve value of E and voltage drop across R_1 is equal to 0.5 V.

Example 5.25. Determine V_o and required PIV rating of each of the diodes of the following circuit (Fig. 5.35).

[U.P. Technical Univ. Electronics Engineering First Semester, 2006-07]

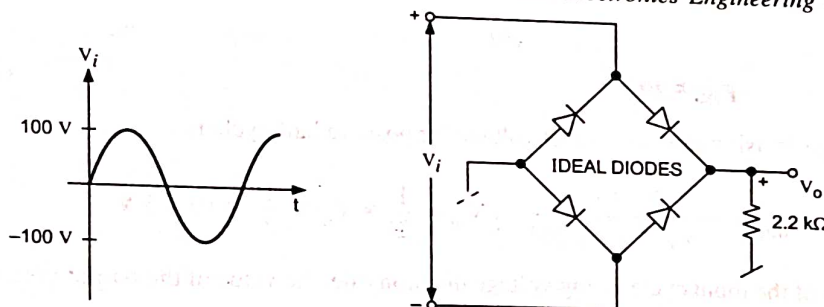


Fig. 5.35

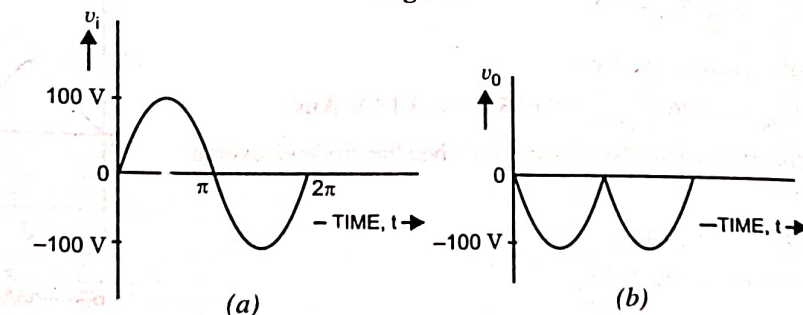


Fig. 5.36

Solution:

$$\text{Rectified output, } V_{\text{dc}} = \frac{2 V_{s \text{ max}}}{\pi} = \frac{2 \times 100}{\pi} = 63.66 \text{ V Ans.}$$

$$\text{PIV rating of each diode} = V_{s \text{ max}} = 100 \text{ V Ans.}$$

Example 5.26. Determine the output waveform of the network shown in Fig. 5.37 and calculate the required PIV of each diode.

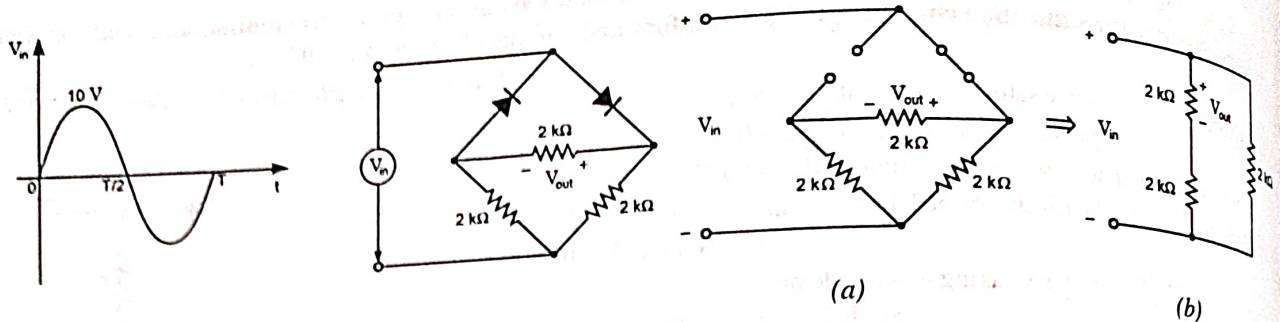


Fig. 5.37

Fig. 5.38

Solution: For the positive input and negative input half cycles, the circuit will appear as shown in Figs. 5.38 and 5.39 respectively.

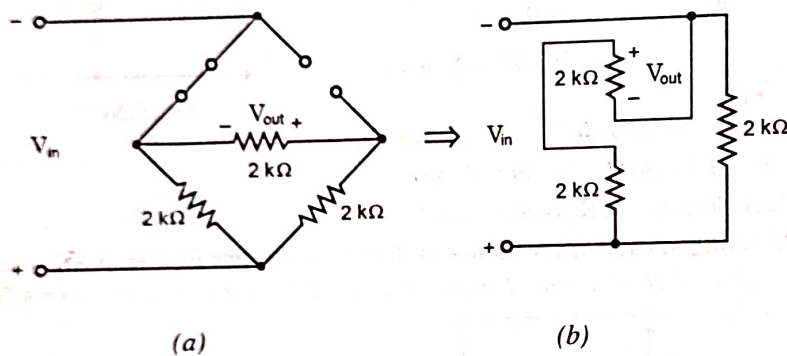


Fig. 5.39

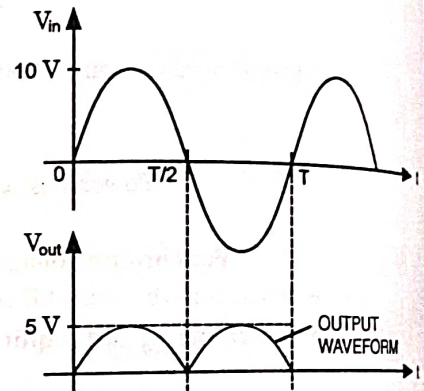


Fig. 5.40

Now, using voltage division rule, the output voltage for positive half cycle is

$$V_{out} = \frac{2 \times 10^3}{2 \times 10^3 + 2 \times 10^3} \times V_{in} = \frac{1}{2} \times V_{in} = \frac{1}{2} \times 10 = 5 \text{ V}$$

For the (-) ve part of the input cycle, using voltage division rule, the value of the output voltage will be

$$V_{out} = \frac{1}{2} V_{in} = 5 \text{ V}$$

Now, from the output waveform we have

$$V_{dc} = 0.636 \times V_{out} = 0.636 \times 5 = 3.18 \text{ V Ans.}$$

PIV is the voltage appearing across the open circuit when the diode is reverse biased. In this case, we have

$$\text{PIV} = 5 \text{ V Ans.}$$

Output waveform is shown in Fig. 5.40.

Example 5.27. Determine V_{out} for the network given in Fig. 5.41.

Solution: From the direction of the diode and the polarity of the dc supply of 2 V, it can be observed that the diode will be in the 'ON' state for a part of the +ve region of the input signal.

The diode will be ON for those inputs satisfying the condition $V_{in} < 2 \text{ V}$.

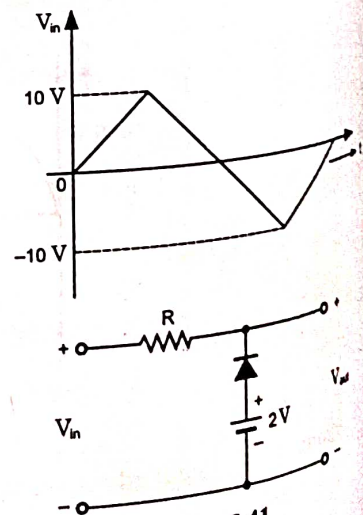


Fig. 5.41

Thus, the transition state of the diode is $V_{in} = 2 \text{ V}$.

When $V_{in} < 2 \text{ V}$, the diode is ON and the output will appear as shown in Fig. 5.42(c). Hence $V_{out} = 2 \text{ V}$.

When $V_{in} > 2 \text{ V}$ the diode is OFF and the output will appear as shown in Fig. 5.42(c) where $V_{out} = V_{in}$ Ans.

During negative half cycle diode will be ON and, therefore, output voltage $V_{out} = 2 \text{ V}$, as shown in the Fig. 5.42(c).

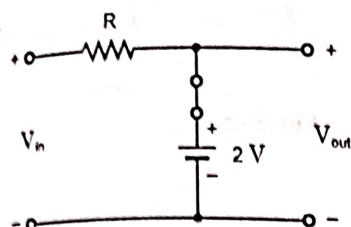


Fig. 5.42(a)

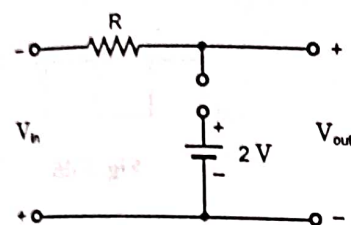


Fig. 5.42(b)

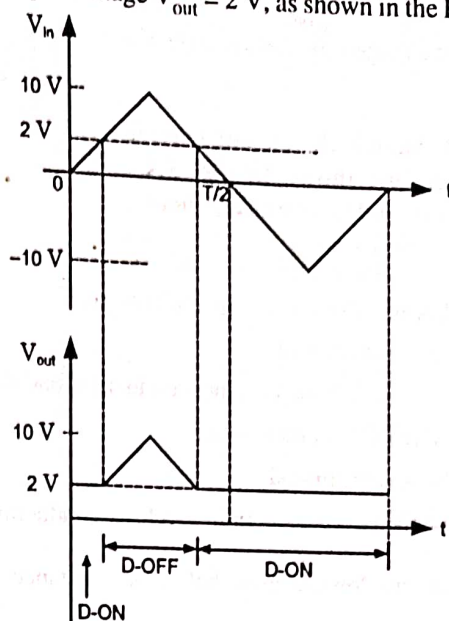


Fig. 5.42(c)

Example 5.28. A symmetrical 5 kHz square wave that varies between +10 V and -10 V is impressed upon the clipping circuit shown in Fig. 5.43. Assume $R_f = 0$, $R_r = 2 \text{ M}\Omega$ and $V_j = 0$. Sketch the steady output waveform, indicating values of the maximum, minimum and constant portions.

[U.P.S.C. I.E.S. Electrical Engineering-II, 2006]

Solution: During positive half cycle of the input, the condition of the circuit is depicted in Fig. 5.44(a). During this period, diode remains reverse biased hence replaced by $R_r = 2 \text{ M}\Omega$. If the current flowing through the input loop is I , then applying KVL to this loop we have

$$2 \times 10^6 I + 1 \times 10^6 \times I + 2.5 = V_{in} = 10 \text{ V}$$

$$\text{or } I = \frac{10 - 2.5}{3 \times 10^6} = 2.5 \times 10^{-6} \text{ A}$$

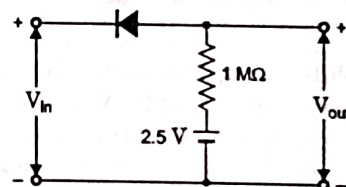


Fig. 5.43

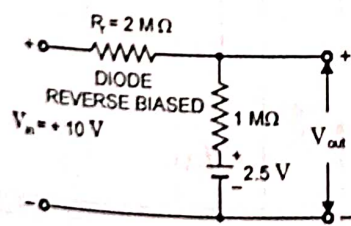


Fig. 5.44(a)

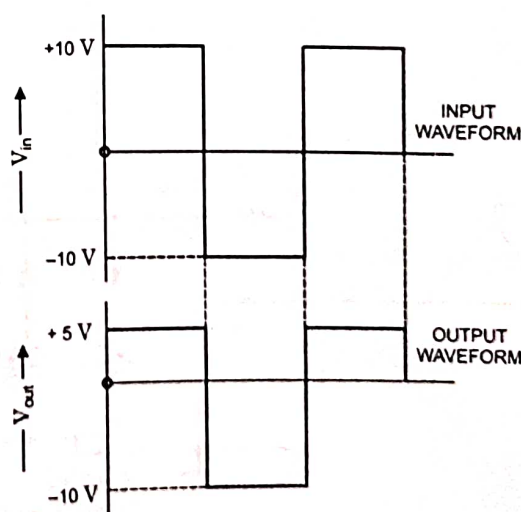


Fig. 5.44(b)

$$\text{Output voltage, } V_{\text{out}} = I \times 1 \times 10^6 + 2.5 = 2.5 \times 10^{-6} \times 1 \times 10^6 + 2.5 = 5 \text{ V}$$

During negative half cycle of the input the diode remains forward biased and hence the output voltage

$$V_{\text{out}} = V_{\text{in}} = -10 \text{ V}$$

The output voltage waveform is shown in Fig. 5.44 (b).

Example 5.29. Sketch the output voltage V_{out} for the circuit shown in Fig. 5.45. Assume diodes D_1 and D_2 to be ideal diodes.

[U.P. Technical Univ. Basic Electronics
Second Semester, 2003-04]

Solution: (a) For positive half cycle of input wave,

- (1) D_1 will be forward biased,
- (2) D_2 will be reverse biased i.e., not conducting, and therefore, output will be zero.

(b) For negative cycle of input wave

- (1) D_1 will be reverse biased,
- (2) D_2 will be forward biased and, therefore, conducting.

$$\text{The current flowing through the load resistance, } I_L = \frac{v_{\text{in}}}{(5 + 5) \times 10^3}$$

$$\text{and output voltage, } v_{\text{out}} = I_L \times R_L = I_L \times 5 \times 10^3 = \frac{v_{\text{in}}}{10 \times 10^3} \times 5 \times 10^3 = \frac{v_{\text{in}}}{2}$$

Output waveform is shown in Fig. 5.46.

Example 5.30. For the circuit shown in Fig. 5.47 sketch i_R and (D_1 and D_2 are Si diode.)

[U.P. Technical Univ. Electronic Engineering Second Semester, 2007-08]
Solution: Since both of the diodes are of silicon, forward voltage drop across each of them will be 0.7 V. For the positive input half cycle, diode D_2 will be reverse biased and diode D_1 will conduct when $v_{\text{in}} \geq 5.3 + 0.7 = 6 \text{ V}$.

For the negative half cycle, diode D_1 will remain reverse biased and diode D_2 will conduct when $v_{\text{in}} \leq -7.3 + 0.8 = 8 \text{ V}$.

The input and output and i_R waveforms are shown in Fig. 5.48.

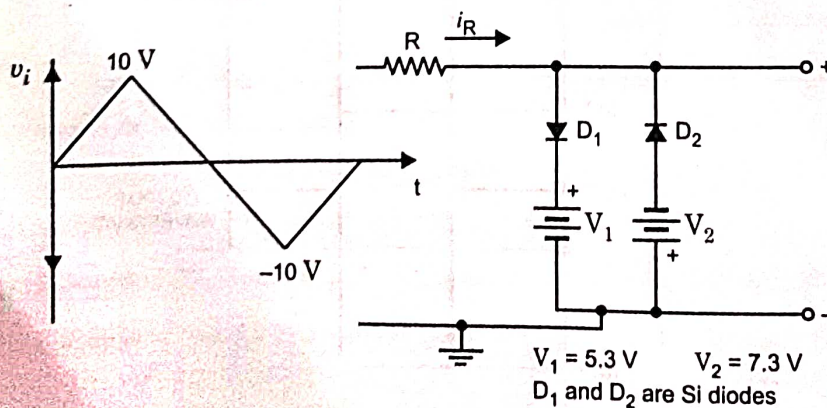
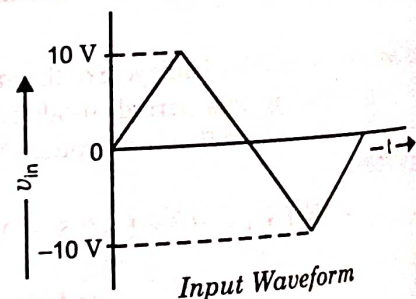
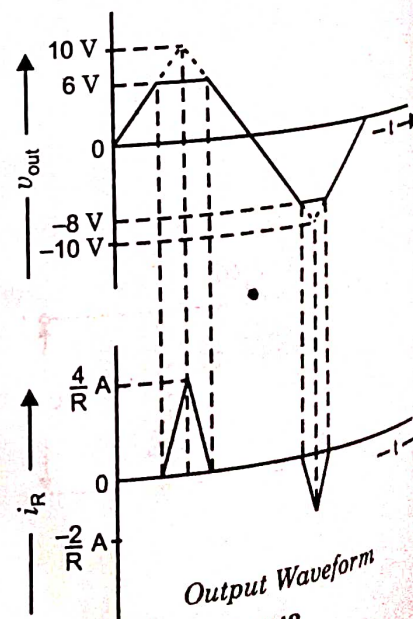


Fig. 5.47



Input Waveform



Output Waveform
Fig. 5.48

Example 5.31. Draw the output waveform for the circuit shown in Fig. 5.49 for the given input waveform.

[U.P. Technical Univ. Electronics Engineering Second Semester 2005-06]

Solution: Assuming ideal diodes i.e., $V_F = 0$

For Diode D_1 . During positive half cycle : Diode D_1 remains reverse biased till $v_{in} < 10$ V, so $v_{01} = v_{in}$ and when $v_{in} > 10$ V, diode D_1 becomes forward biased and $v_{01} = 10$ V

Again when $V_{in} < 10$ V, D_1 is reverse biased and, therefore, $v_{01} = v_{in}$

During negative half cycle : Diode D_1 remains reverse biased and $v_{01} = v_{in}$

The waveform for v_{01} is illustrated in Fig. 5.50 (b).

For Diode D_2 . During positive half cycle : When $V_{in} < 5$ V, Diode D_2 remains forward biased. So $v_{02} = 5$ V.

When $V_{in} > 5$ V, diode D_2 becomes reverse biased and $v_{out} = v_{in}$

Again when $v_{in} < 5$ V, D_2 is forward biased and $v_{02} = 5$ V.

During negative half cycle diode D_2 is forward biased and, therefore, $v_{02} = 5$ V.

The waveform for v_{02} is shown in Fig. 5.50 (c).

Thus during positive half cycle $v_{out} = 5$ V, till $v_{in} = 5$ V.

When $v_{in} > 5$ V, $v_{out} = v_{01}$

Again when $v_{in} < 5$ V, $v_{out} = 5$ V. During negative half cycle,

$$v_{out} = v_{02} = 5 \text{ V.}$$

This is illustrated in Fig. 5.50 (d).

Example 5.32. For the circuit shown in Fig. 5.51, find and sketch the output voltage $v_o(t)$ for $0 \leq t \leq 5$ ms. Assume that the diode is ideal.

[G.G.S.I.P. Univ. Analog Electronics, December-2010]

Solution: Let the input voltage be expressed as

$$v_{in}(t) = Kt \text{ where } t \text{ is in ms}$$

Substituting $v_{in}(t) = 10$ V and $t = 5$ ms, we have

$$k = \frac{10}{5} = 2 \text{ V/ms}$$

$$\text{So } v_{in}(t) = 2t \quad \dots(i)$$

The semiconductor diode remains reverse biased as long as input voltage does not exceed 6 V, so

$$\text{Output voltage, } v_{out}(t) = 6 \text{ V}$$

$$\text{for } v_{in}(t) \leq 6 \text{ V}$$

$$\text{or } t \leq \frac{6}{2} \text{ ms} \leq 3 \text{ ms}$$

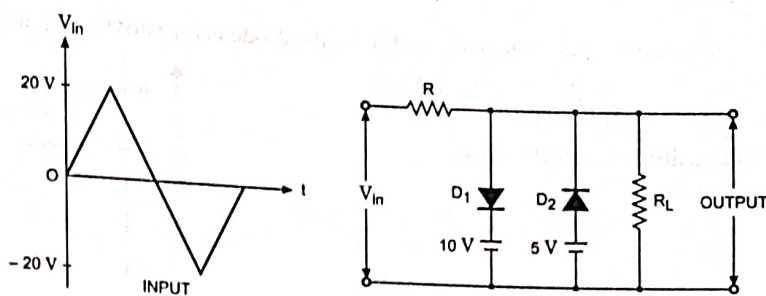


Fig. 5.49

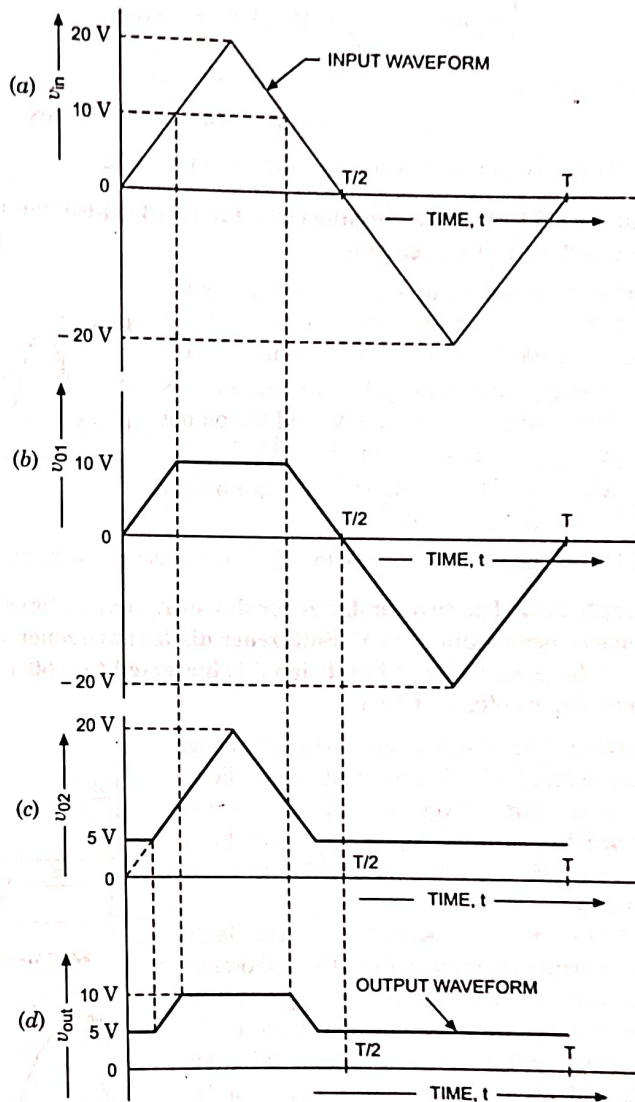


Fig. 5.50

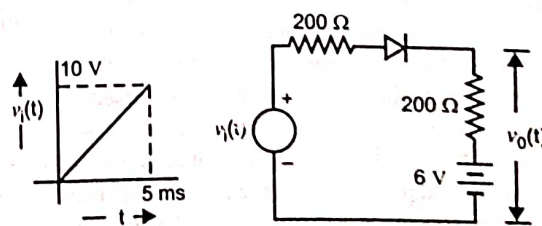


Fig. 5.51

When the input voltage exceeds 6 V, the diode is forward biased and current supplied is

$$= \frac{v_{in}(t) - 6}{200 + 200} = \frac{v_{in}(t) - 6}{400}$$

Voltage drop in 200 Ω resistor

$$= \frac{v_{in}(t) - 6}{400} \times 200 = \frac{1}{2} v_{in}(t) - 3$$

$$\text{So } v_{out}(t) = v_{in}(t) - \left[\frac{1}{2} v_{in}(t) - 3 \right]$$

$$= \frac{1}{2} v_{in}(t) + 3 = \frac{1}{2} \times 2t + 3 = t + 3 \text{ volts}$$

So output voltage $v_{out}(t) = 6 \text{ V}$ for $t \leq 3 \text{ ms}$
and $v_{out}(t) = t + 3 \text{ volts}$ for $3 \text{ ms} \leq t \leq 5 \text{ ms}$

Sketch of output waveform is shown in Fig. 5.52.

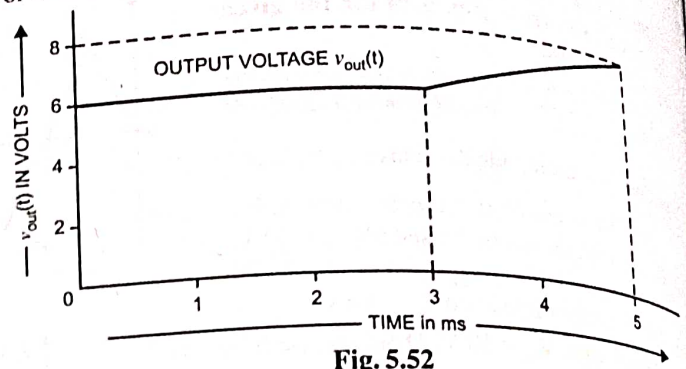


Fig. 5.52

Example 5.33. Design a clipping circuit to clip a sinusoidal wave of 2 V peak voltage at 100 Hz above +1 volt. Draw input and output waveforms.
[G.G.S.I.P. Univ. Analog Electronics-II, May-June-2009]

Solution: Biased positive clipper circuit shown in Fig. 5.53 will clip the sinusoidal wave of 2 V peak volt at 100 Hz above +1 V, as detailed below:

During positive half cycle, the diode remains reverse biased for $0 < v_{in} < 1 \text{ V}$, and the output voltage $v_{out} = v_{in}$ as shown in Fig. 5.53.

When $v_{in} \geq 1 \text{ V}$ diode D is forward biased and, therefore, output voltage $v_{out} = 1 \text{ V}$.

During negative half cycle the diode remains reverse biased and therefore, $v_{out} = v_{in}$ as shown in Fig. 5.53.

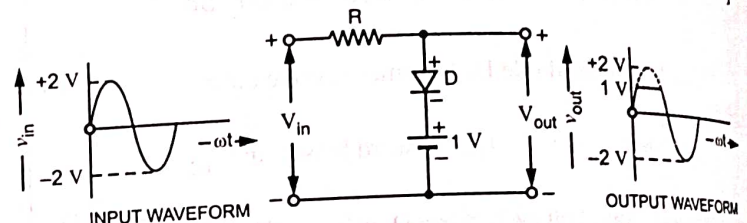


Fig. 5.53

Example 5.34. The two similar zener diodes Z_1 and Z_2 have been joined back-to-back across the input sinusoidal voltage of peak value $\pm 25 \text{ V}$. Both zener diodes have zener voltage of 20 V. Find the output waveform.

If the peak value of input signal is increased to $\pm 60 \text{ V}$ and zener diodes are replaced by zener diodes each having zener voltage of 10 V.

Solution: The given circuit will appear as shown in Fig. 5.54. We know that a zener diode acts like a 'short' (or very low resistance) when forward biased and 'open' (or very high resistance) when reverse biased till it goes into breakdown at V_Z .

During positive input half cycle, zener diode Z_1 is forward biased and so acts like a short but zener diode Z_2 acts like an open up to 20 V. Thereafter Z_2 goes into breakdown and holds the output voltage constant at 20 V till input voltage falls below 20 V in the later part of the positive half cycle. When the supply voltage falls below 20 V, the diode Z_2 comes out of the breakdown and again acts an 'open' across the load circuit. During negative input half cycle, the action of zener diodes Z_1 and Z_2 are reversed. The result is that the output voltage waveform is clipped off on both peaks, as illustrated in Fig. 5.54 (a).

When the peak value of the input signal is increased to 60 V and zener diodes of lesser zener voltage (10 V) are used, the circuit appears as shown in Fig. 5.54 (b) and the output waveform obtained is almost of square shape.

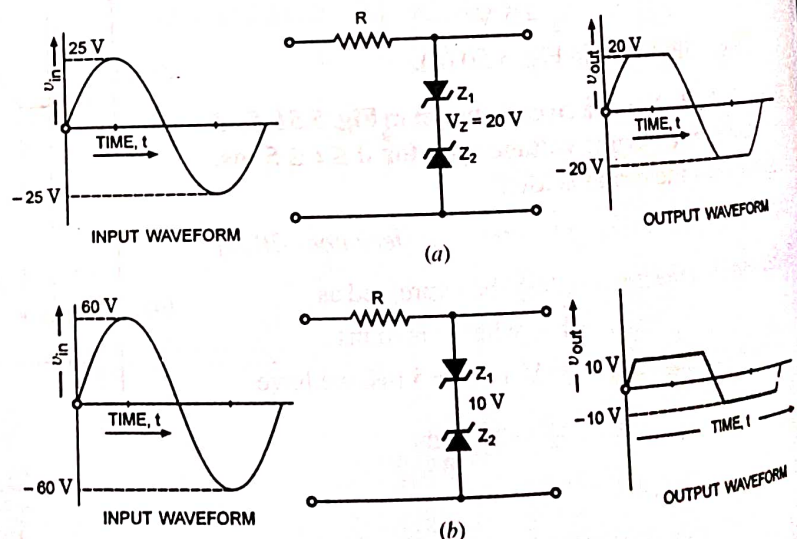


Fig. 5.54

5.9 DIODE CLAMPING CIRCUITS

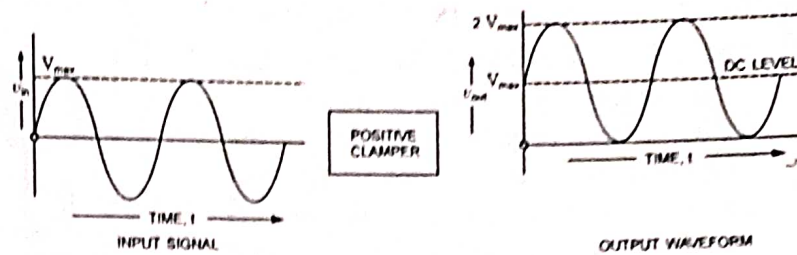
A circuit that places either the +ve or -ve peak of a signal at a desired level is known as a *clamping circuit*. It simply adds or subtracts a dc component to the input signal. In steady state, the output waveform is an exact replica of the input waveform. The clumper is also referred to as a *dc restorer* and a (signal) *level shifter*.

The necessity to establish the extremity of the positive or negative signal excursion at some reference level (i.e., to introduce a dc component to the input signal) arises in connection with a signal that has passed through a capacitive coupling network and lost its dc component, as in case of television receivers. However, normally the dc component introduced is not identical with the dc component lost in transmission.

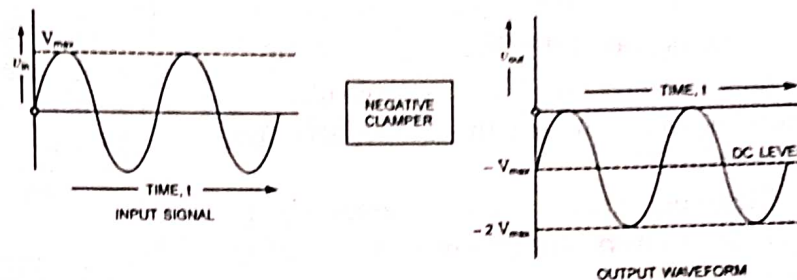
In fact, a clamp circuit adds dc component (positive or negative) to the input signal so as to push it either on the positive side, as illustrated in Fig. 5.55 (a) or on the negative side, as illustrated in Fig. 5.55 (b). When the circuit pushes the signal on the positive side or upward, the negative peak of the signal coincides with the zero* level and the circuit is called the *positive clamper*. On the other hand, when the signal is pushed on the negative side or downward, as illustrated in Fig. 5.55 (b), the positive peak of the input signal coincides with the zero level and the circuit is called the *negative clamper*.

For a clamping circuit at least three components—a diode, a capacitor and a resistor are required. Sometimes an independent dc supply is also required to cause an additional shift. The important points regarding clamping circuits are :

- Waveform remains the same but its level is shifted either upward or downward.
- The clamping circuit does not change the peak-to-peak or rms value of the waveform. Thus referring to Fig. 5.55, the input waveform and output waveform have the same peak-to-peak value i.e., $2 V_{\max}$. If the input voltage and clamped output are measured by an ac voltmeter, the readings will be the same.
- The clamping circuit, of course, alters the peak and average values of the waveform. In Fig. 5.55 the input waveform has a peak value of $V_{\max}$ and average value over a complete



(a) Positive Clamping



(b) Negative Clamping

Fig. 5.55

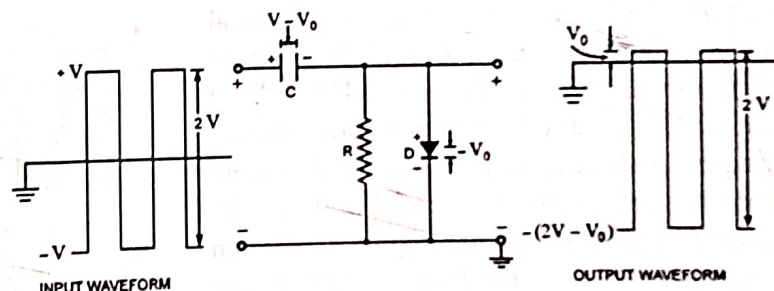


Fig. 5.56 Negative Clamper

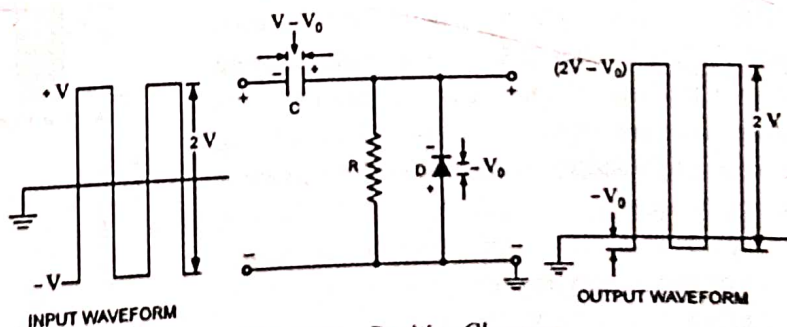


Fig. 5.57 Positive Clamper

* Assuming diode to be an ideal one.

Example 5.35. Sketch V_o for the network shown in Fig. 5.59.

[U.P. Technical Univ. Electronics Engineering Second Semester 2006-07; Mayamaya Technical Univ. Second Semester 2010-11]

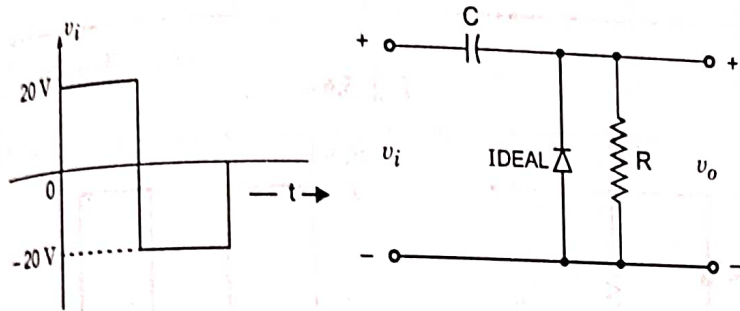


Fig. 5.59

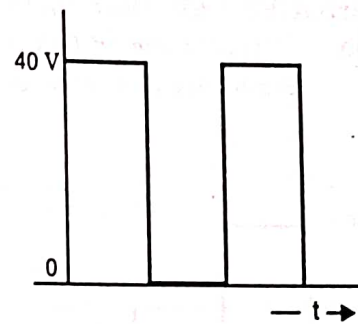


Fig. 5.60

Solution: Given circuit is of positive clamper. In positive half cycle, capacitor will shift the voltage to positive direction and output voltage will be $2 \times V_{in} = 40 \text{ V}$.

During negative half cycle, diode will be forward biased and there is no voltage drop across diode (diode being ideal) and therefore output voltage will be zero. Output waveform is shown in Fig. 5.60.

Example 5.36. Sketch V_o for the network of Fig. 5.61 for the input shown.

[Mahamaya Technical Univ. Electronics Engineering Second Semester 2010-11]

Solution: Given circuit is of positive clamper.

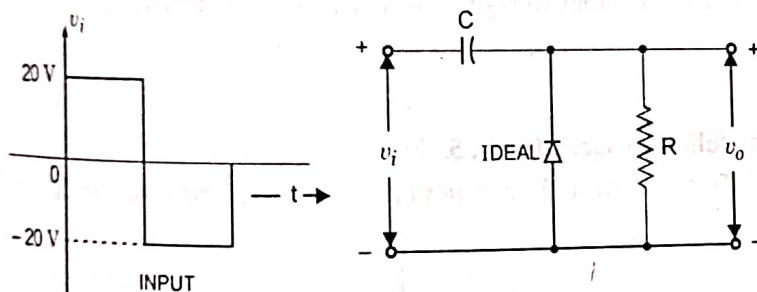


Fig. 5.61

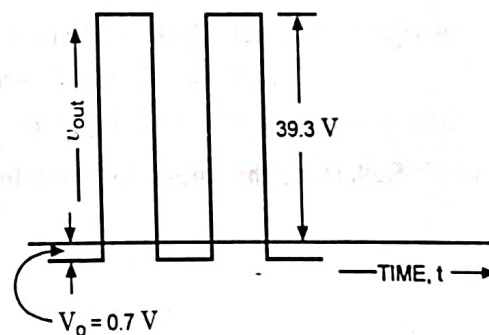


Fig. 5.62

In positive half cycle capacitor will shift the voltage to positive direction and output voltage will be $V_{out} = 2V_{in} - V_F = (2 \times 20 - 0.7) \text{ V} = 39.3 \text{ V}$ assuming silicon diode.

During negative half cycle, the diode will be forward biased and output voltage will be equal to the forward drop of diode i.e., 0.7 V .

The output waveform is shown in Fig. 5.62.

Example 5.37. Sketch out voltage (V_o) waveform of the following network as shown in Fig. 5.63. Assume diode is silicon with $V_0 = 0.7$ V. [G.B. Technical Univ. Electronics Engineering First Semester, 2012-13]

Solution: Given circuit is of a positive clamper. During negative half cycle of the input, the output voltage is equal to battery voltage less barrier potential of diode i.e. $(5 - 0.7)$ V or 4.3 V.

During positive output, the diode D is forward biased and the capacitor is charged with the polarity shown. The capacitor C is charged to $(20 + 5 - 0.7)$ i.e. 24.3 V.

During positive input, the output voltage being the sum of the positive input voltage and capacitor voltage i.e. $10 + 24.3$ i.e. 34.3 V.

Output waveform is shown in Fig. 5.64.

Example 5.38. Determine V_o for the following circuit (Fig. 5.65). Also name the configuration. [U.P. Technical Univ. Electronics Engineering First Semester 2006-07]

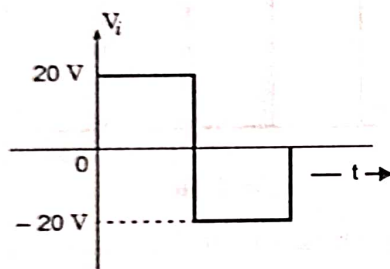


Fig. 5.65

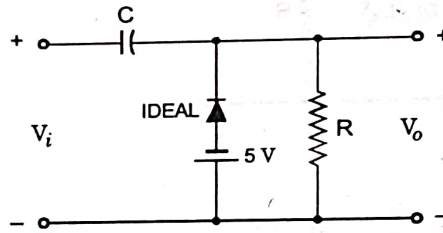


Fig. 5.64

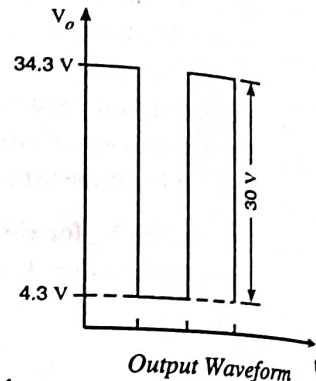


Fig. 5.66 Output Waveform

Solution: The given circuit is of positive clamper. In positive half cycle, capacitor will shift the voltage in positive direction, but battery will be in negative direction.

$$\text{So output voltage } V_o = 2V_i - V_1 = 2 \times 20 - 5 = 35 \text{ V}$$

In negative half cycle, diode will be forward biased and output voltage V_o will be equal to battery voltage

$$\text{i.e., } V_o = -V_1 = -5 \text{ V Ans.}$$

Output waveform is shown in Fig. 5.66.

Example 5.39. Draw the output waveform for the following circuit (Fig. 5.67).

[U.P. Technical Univ. Electronics Engineering Second Semester, 2004-05]

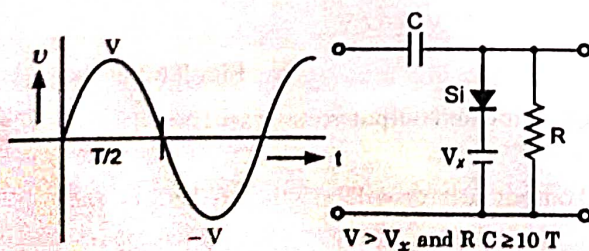


Fig. 5.67

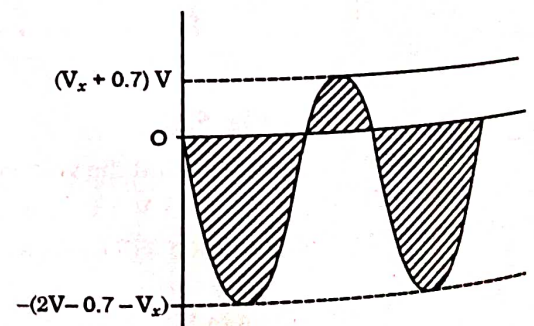


Fig. 5.68

Solution: Given circuit is of negative clamper.
Output waveform is shown in Fig. 5.68.

Example 5.40. Design a clamper to perform the function indicated in Fig. 5.69.

[G.B. Technical Univ. Electronics Engineering First Semester 2011-12]

Solution: The circuit to perform the function indicated in Fig. 5.69 is given in Fig. 5.70.

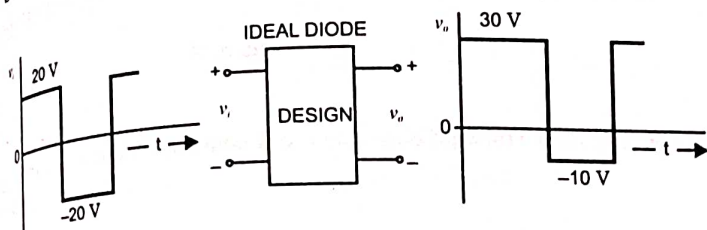


Fig. 5.69

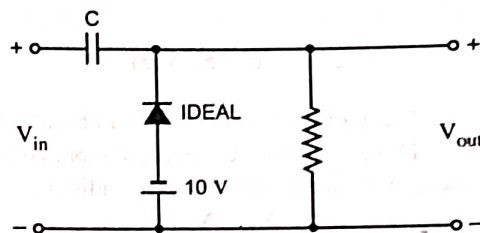


Fig. 5.70

In positive half cycle, the capacitor will shift the voltage in positive direction, but the battery will be in negative direction. So, output voltage $V_{out} = 2V_{in} - V = 2 \times 20 - 10 = 30 \text{ V}$

In negative half cycle, the diode will be forward biased and the output voltage will be equal to battery voltage i.e., 10 V.

Note: Battery voltage will be equal to the output voltage during negative half cycle.

Example 5.41. Design a clamper to perform the function indicated in Fig. 5.71.

[G.B. Technical Univ. Electronics Engineering First Semester 2012-13]

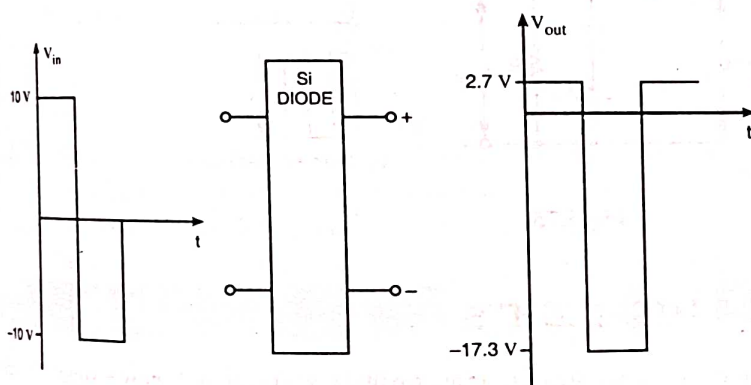


Fig. 5.71

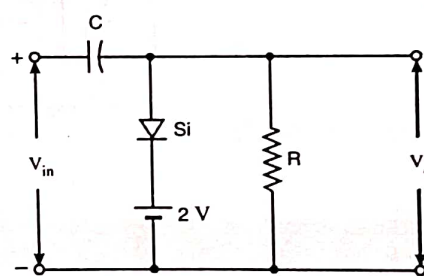


Fig. 5.72

Solution: The circuit to perform the function indicated in Fig. 5.71 is given in Fig. 5.72. During positive half cycle the diode is forward biased and the output voltage, $V_{out} = 2 + 0.7 = 2.7 \text{ V}$. During negative half cycle, the capacitor will shift the voltage in negative direction but the battery will be in positive direction so the output voltage, $V_{out} = -(-2 \text{ V} - V_o - \text{battery voltage}) = -(-2 \times 10 - 0.7 - 2) = -17.3 \text{ V}$.

Example 5.42. For the circuit shown in Fig. 5.73 (a) and Fig. 5.73 (b), sketch and explain the output waveforms. Assume the diodes to be ideal.

[U.P.S.C. I.E.S. E&T Engineering-II, 2006]

Solution: During positive half cycles, the diode remains reverse biased for $0 < V_{in} < 2 \text{ V}$, and the output voltage $v_{out} = v_{in}$

When $V_{in} \geq 2 \text{ V}$ diode D is forward biased and, therefore, conducts and output voltage $v_{out} = 2 \text{ V}$

During negative half cycle the diode D remains reverse biased and therefore $v_{out} = v_{in}$

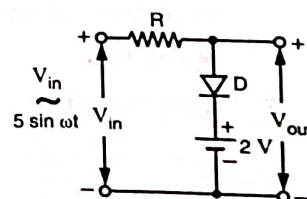


Fig. 5.73 (a)

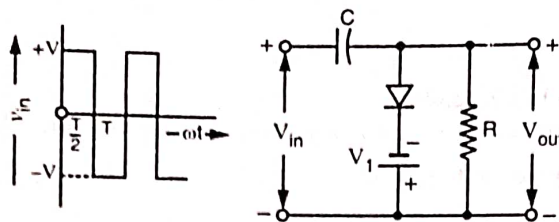


Fig. 5.73 (b)

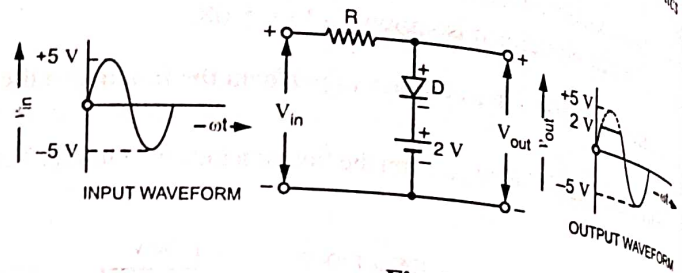


Fig. 5.74

Output waveform is shown in Fig. 5.74.

During positive half cycle of input voltage, the diode remains forward biased and so conducts and the capacitor C gets charged and under steady-state condition

$$V_{in} - V_C + V_1 = 0$$

or Capacitor voltage, $V_C = V_{in} + V_1 = V + V_1$

and output voltage, $V_{out} = -V_1$

During negative half cycle of input voltage, diode D remains reverse biased and so does not conduct.

$$\text{So, } -V - V_C - V_{out} = 0$$

$$\text{or } V_{out} = -(V + V_C) = -(V + V + V_1) = -(2V + V_1)$$

Output waveform is shown in Fig. 5.75.

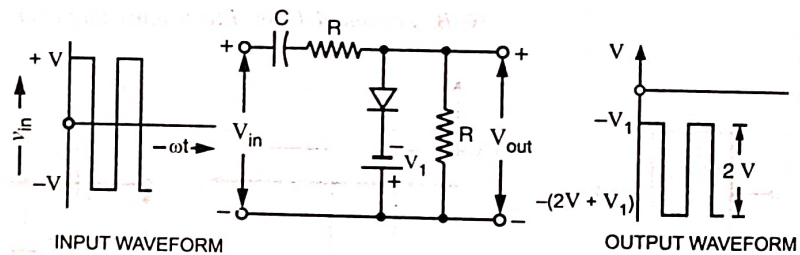


Fig. 5.75